

Machine Learning-Based Estimation of Ripple Losses in High Temperature Superconductors for Cryo-Electric Aviation Applications

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Abstract—Losses generated by high temperature superconducting (HTS) tapes are a source of heat load in the powertrain of superconducting driven electric aircraft. This study employs machine learning (ML) techniques to estimate ripple losses of HTS tapes, which occur on the DC side of the powertrain and place notable demands on the cryogenic cooling system. Ripple losses generated by 6, 12, and 24 pole (pulse) rectifications were investigated under switching frequencies of typical AC/DC converters in electric aircraft. Fast, data driven estimation of ripple losses in the HTS tapes are introduced to overcome the high computational cost of FEM-based methods. Four ML techniques, Gaussian process regression (GPR), decision tree (DT), ensemble tree (ENS), and artificial neural network (ANN), are assessed to demonstrate the effectiveness of the best method. As current ripple may propagate through the full drivetrain, the choice of ML technique is a technical requirement, and rapid, tape level loss prediction is a primary step for condition monitoring and health assessment of HTS devices in cryo-electric aircraft. Training data were generated using a validated 2D H-formulation finite element method (FEM)-based model developed in COMSOL Multiphysics, which included all metallic sublayers to account for their high frequency effects. Results showed GPR achieves the highest accuracy, with a mean relative error of 1.8% and goodness of fit percentage value of 99.997%. The model is then updated to a two-tape stack model to show the generalizability of the proposed GPR model, also showing high accuracy results. ML techniques reduced computation time dramatically compared with FEM simulation, with test times between 51 ms and 139 ms for the full data range, compared to 10–30 minutes for only the frequency range data.

Index Terms—Artificial intelligence, Current ripple, Electric aircraft, Loss prediction, Superconducting propulsion.

I. INTRODUCTION

Globally, aviation is responsible for around 2%-4% of CO₂ emissions [1-3]. As part of efforts to reduce this, technologies such as superconductivity and hydrogen (cryo-electrification) have been explored as key enablers for clean aviation [4]. Second generation (2G) high temperature superconducting (HTS) tapes are key components of HTS devices [5]. The benefits of superconducting drivetrains have been demonstrated in programs such as Airbus UpNext's ASCEND (Advanced Superconducting Cryogenic Experimental Powertrain Demonstrator – 2021 to 2023) [6], the EU funded ASuMED (Advanced Superconducting Motor

Experimental Demonstrator – 2017 to 2020) program [7], Airbus' Cryo-Prop project (2024 to 2027) [8], and the NASA backed CHEETA (Centre for High-Efficiency Electrical Technologies for Aircraft – 2019 to 2024) program [9].

Loss is a key factor for monitoring the efficiency of powertrains and their heat load. Due to the cryogenic operation of HTS devices, a loss of 1 W at 77 K may translate to a cooling power requirement equivalent to 12 W – 20 W at room temperature [10]. This large increase in loss is made worse at lower temperatures, with 1 W at 20 K translating to above 100 W at room temperature [11, 12]. Current discussion within the research community and industry shows that there is a direction towards lower temperatures using hydrogen as a fuel source and additionally as a heat sink for achieving 'almost free cooling', hence reducing the required capacity from the cryocooler, as discussed in [13] and [14]. Hence, if higher loss is observed at lower temperatures, swift prediction of these loss magnitudes are key so the recovery system can actively deal with them is.

In the context of aviation, cryo-electric refers to aircraft utilizing superconducting or cryogenic engineering in the power network or propulsion system for high efficiency, low loss and reduced carbon footprint. During the rectification of AC power to DC power, small current distortions appear on the DC current signal, known as ripples [15, 16]. The nonlinear resistive behavior of HTS tapes makes them sensitive to such disturbances.

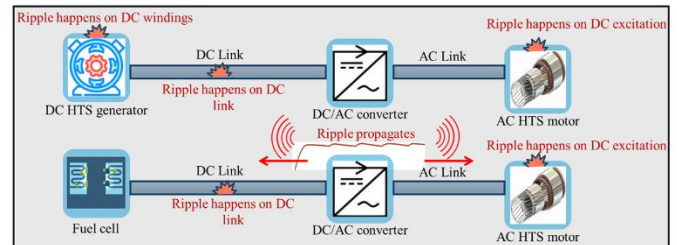


Fig. 1 Typical drive train for a cryo-electric aircraft depicting where a superimposed ripple may manifest

This study is particularly relevant to DC power systems in which current ripples originating from power electronic converters propagate through the power train and interact with superconducting components, as shown in Fig. 1. In weight and cooling limited aircraft systems, this can reduce efficiency and

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risk thermal instability or local heating/quenching [17, 18]. In these systems, understanding and quantifying ripple-induced losses is essential for cooling system design, reliable operation of the power train, and development of effective fault detection and recovery systems.

Previous research studies focused on loss estimation and characterization caused by harmonics [19-23], as well as characterizing ripple losses experimentally [17, 24, 25], and using Finite Element Method (FEM) [26-29]. FEM-based models typically have long simulation time and high computational costs for complex geometries, and even at tape level can be 10 to 30 minutes depending on the size and density of the applied mesh. Thus, they are impractical for system level analysis, real-time simulation of HTS devices, and online condition monitoring of cryo-electric aircraft.

In comparison with FEM based models, Artificial Intelligence (AI) and Machine Learning (ML) algorithms (which are a subset of AI techniques) offer fast, flexible, and easily adaptable modelling approaches (while maintaining high precision), reducing computational cost and time for complex problems [30]. Recent studies have applied ML for efficient surrogate modelling of HTS tapes and devices. These surrogate models approximate complex system behavior without directly solving the governing equations [31]. ML based techniques have been used for estimation of various parameters, including temperature, of superconducting fault current limiters [32-34], short circuit characteristics of HTS cables [35], loss estimation in superconducting devices [36-38], and critical current and force estimation of superconductors [39-42].

Until now, ripple type losses in HTS tapes and devices have been estimated using numerical FEM-based models. These models are computationally demanding due to the high-frequency content of ripples found in electric propulsion systems, which necessitates fine sublayer resolution within the HTS tape, unlike low-frequency AC loss calculations. Although FEM-based methods for ripple loss estimation are well established, fast and data-driven approaches have not yet been explored. Rapid, tape-level estimation of ripple losses is crucial for the condition monitoring of HTS devices in cryo-electric aircraft, where such tapes form the basic building blocks of larger HTS components in propulsion motors.

While external faults such as short circuits influence overall machine performance, localized incipient anomalies affect the electromagnetic behaviour of individual tapes. Hence, the behavior of individual tapes will directly govern the electromagnetic performance the machines. Tape-level analysis provides the necessary understanding to capture localized degradation mechanisms or incipient anomalies that may eventually lead to failures under operational stresses. An incipient fault develops at an early stage and gradually alters the electromagnetic behavior of the tape without immediately causing failure. A soft fault is an internal anomaly in an HTS tape or coil that affects performance but does not trigger an immediate system shutdown.

Integrating such fast, tape-level predictive tools into real-time monitoring frameworks can significantly enhance the reliability and operational safety of cryo-electric propulsion systems and

therefore a gap exists in literature for efficient ripple loss estimation methods capable of capturing tape-level behaviour to support early detection of faults to ensure HTS machine reliability.

In this paper, for the first time, ML-based techniques are used for fast ripple loss estimation in HTS ReBCO tapes. The loss performance and behavior of an HTS tape are simulated using COMSOL Multiphysics under the influence of a ripple current superimposed on a DC current. Loss data is extracted for 6, 12 and 24 pole rectification at a range of frequencies and at varying levels of DC transport current [15, 16]. In this context, the term pole typically refers to the positive (or negative) peak in the ripple waveform of a DC output resulting from the rectification process. These are also sometimes referred to as pulses. Then, the losses are estimated using ML based intelligent regressor algorithms, namely, Gaussian process regression (GPR), decision tree (DT), ensemble and artificial neural networks (ANN), allowing for timely calculation of ripple losses. It has been shown that the GPR regressor showed the highest accuracy for loss estimation, with the computation time within the millisecond range, compared to 10-30 minutes for FEM models under basic frequency parameter sweeps, without even accounting for pole number or transport current variation. Furthermore, commonly used 10 mm and 12 mm tape widths, as well as different operating temperatures, are considered to show the robustness of the proposed regressor. Finally, the model is adapted to a two-tape stack model to enforce the generalizability of the proposed GPR regressor, again showing positive results for low relative error and a positive goodness of fit.

II. NUMERICAL MODELLING

A. Electromagnetic modelling

The loss of an HTS tape carrying DC current with ripples is modelled using a 2D H-formulation FEM based model, where the parallel and perpendicular magnetic field components are considered as the dependent variables [43].

The Ampere's and Faraday's Law derivation that have been applied to the H-formulation model are shown in Equations (1) and (2), respectively [43]:

$$\mathbf{J} = \nabla \times \mathbf{H} \quad (1)$$

$$\nabla \times (\rho \nabla \times \mathbf{H}) = - \frac{\partial(\mu_0 \mu_r \mathbf{H})}{\partial t} \quad (2)$$

where μ_0 is vacuum permeability and μ_r is relative permeability. It should be noted that $\mathbf{H}$ is the magnetic field strength, $\mathbf{J}$ is the current density, and ρ is resistivity.

The well-known $\mathbf{E}\text{-}\mathbf{J}$ power law, shown in Equation (3), has been used to characterize the nonlinear behavior of the electric field ($\mathbf{E}$), as a function of current density ($\mathbf{J}$), and is used to calculate the resistivity of the superconductor [43, 44].

$$\rho(\mathbf{J}) = \frac{E_c}{J_c} \left(\frac{|\mathbf{J}|}{J_c(\mathbf{B})} \right)^{n-1} \quad (3)$$

where $E_c = 10^{-4} \text{ V/m}$, n is the tape index value from the $\mathbf{E}\text{-}\mathbf{J}$

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curve, and $J_c(B)$ is the critical current density dependent on the magnetic field, as shown in Equation (4) [44, 45]:

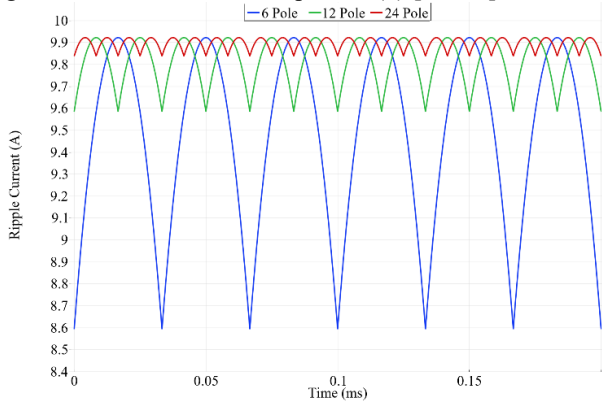


Fig. 2 An arbitrary ripple current waveform with 6, 12, and 24 pole rectification, at 5 kHz frequency for illustration

$$J_c(B) = \frac{J_{c0}}{\left(1 + \frac{\sqrt{\gamma^2 B_{\parallel}^2 + B_{\perp}^2}}{B_0}\right)^{\alpha}} \quad (4)$$

where J_{c0} is the value of critical current density in operational temperature (assumed at 77 K) and self-field and B_0 is a constant determined from the measured E - J curve of coated conductors under perpendicular and parallel magnetic field components, $B_{\perp}$ & $B_{\parallel}$, with $J_{c0} = 5.95 \times 10^4 \text{ A/m}^2$, $B_0 = 42.65 \text{ mT}$, $\gamma = 0.29515$, and $\alpha = 0.7$ [46].

Finally, the ripple loss of the HTS tape is calculated based on Equation (5) [43, 44], with the equation applied separately to each layer of the tape. The total loss is a summation of the loss in each individual layer (Q_n), as in Equation (6) [46]:

$$Q_n = 2 \int_{\frac{T}{2}}^T \int_{\Omega} \mathbf{E} \cdot \mathbf{J} \, d\Omega \, dt \quad (5)$$

$$Q_{tot} = \sum_{n=1}^5 Q_n \quad (6)$$

where t is time in seconds, and T is the fundamental period of one cycle of the applied ripple current.

B. Current Ripple Modelling

To model the ripple current, a p -pulse rectification device is considered, where p is the number of rectifying unit poles; 6, 12, and 24, as used in [15] and [16]. The calculation sampling frequency for the 6, 12 and 24 pole rectification was set large enough to ensure a smooth ripple waveform, hence accurately depicting the electromagnetic characteristics of the ripple losses.

The output of the p -pulse rectification procedure as its Fourier series expansion is shown in (7), where the equation represents a DC current with superimposed time varying ripple current [15].

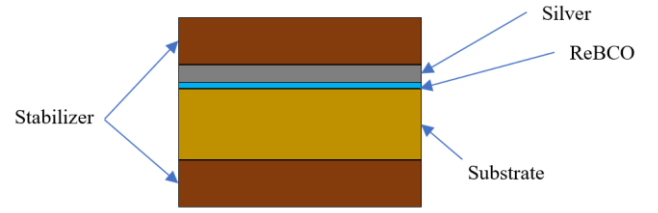


Fig. 3 Structure of the HTS tape showing each layer. Not to scale

$$I_p(t) = I_0 \frac{p}{\pi} \sin\left(\frac{\pi}{p}\right) \left[1 - 2 \sum_{k=1}^{\infty} \frac{(-1)^k}{p^2 k^2 - 1} \cos(pk\omega t) \right] \quad (7)$$

where, I_0 is the amplitude of transport current, and ω is angular frequency in rad/s.

The ripple current waveform shown in **Fig. 2** is obtained. As observed in **Fig. 2** the ripple frequency of the 24-pulse rectification is higher than 12 and 6 pulse rectifications. Hence, the amplitude of the current waveform and its ripple frequency are two important factors which have a direct impact on loss values.

A key choice is the frequency range for the switching frequency. For low frequencies, it is reasonable to simulate only the superconducting layer, as all current flows through this. However, at higher frequencies, some current flows in the substrate and stabilizer layers, hence generating a loss and necessitating simulation of the full tape geometry when considering the proposed use in electric aircraft. In general, electric aircraft are likely to operate within switching frequency ranges of 2 kHz to 200 kHz in their DC part [47-51].

In this study, the HTS tape is modelled as a straight single tape segment, a commonly used simplification as the curved end sections of twisted tapes and coil ends contribute negligibly when the straight length dominates [52, 53]. Moreover, previous comparisons have shown that 2D straight-tape models yield comparable loss predictions to 3D twisted configurations while significantly reducing computational effort [54, 55]. The proposed AI-based loss estimator outputs loss per unit length (J/cycle/m), making it independent of the tape length and directly scalable to coil or cable levels by methods like transfer learning and physic-based surrogate modelling.

The HTS tape, including ReBCO, stabilizer, silver, and substrate layers, is modelled in finite element software COMSOL Multiphysics and solved in the time domain. The structure of the tape is shown in **Fig. 3**. To ensure high accuracy, a mapped distributed mesh is applied to the tape, with a free triangular mesh applied to the air domain that surrounds the tape. The geometry of the tape and parameters used are presented in [46].

III. MACHINE LEARNING MODELLING

While interpolation techniques can approximate intermediate values between known data points, they are fundamentally limited in several ways when compared to ML approaches. Firstly, interpolation methods are typically constrained to low-dimensional problems and cannot efficiently handle multiple

interacting input variables, particularly in highly non-linear problems, as is the case for the coupled electromagnetic dynamics in HTS materials. Secondly, interpolation methods lack adaptability; requiring uniformly sampled grids and reconstruction if additional data are introduced or if the operating range changes.

In contrast, ML-based models are data-driven, scalable, and capable of learning complex non-linear mappings across high-dimensional input spaces without being limited to the training grid. In ML models, each point contributes to the model's understanding of local behaviour, while the entire dataset informs global patterns and correlations. This dual perspective enables ML methods to recognize intricate interdependencies and generalize to new conditions not explicitly represented in training data.

Various ML techniques can estimate losses with differing accuracy and complexity, enabling applications such as real-time cryogenic system monitoring, HTS winding internal fault detection, predictive maintenance, and annual energy loss estimation [30].

This study presents four ML regression techniques: Gaussian process regression (GPR), decision trees (DT), ensemble learning (ENS), and artificial neural networks (ANN). DT, GPR, and ENS perform well with limited data, while ANN excels in complex tasks like ripple loss estimation [56].

A. Gaussian Process Regression

GPR is a non-parametric Bayesian regression method that balances exploitation, refining solutions locally, and exploration, searching new solution regions via larger parameter changes [57].

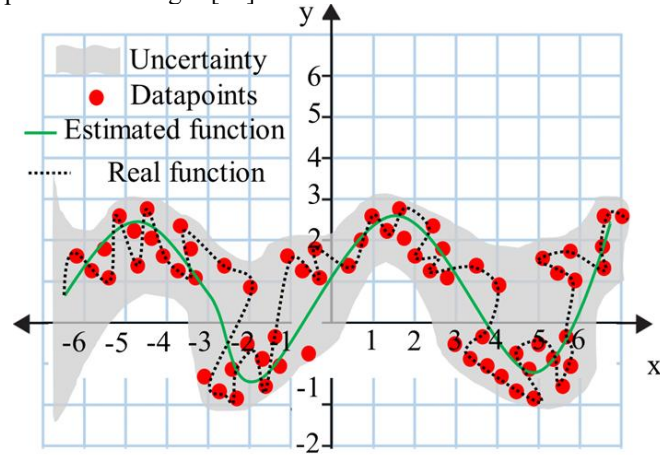


Fig. 4 An example presentation of GPR regressor in 2D, where the grey filler represents the uncertainty, red dots are the datapoints, the green line is the estimated function, and the black dashed line represents the real function

As shown in **Fig. 4**, GPR estimates a function to fit to several datapoints based on their mean value and covariance. Although the estimated function could be different from the real function that governs the datapoints, it can still provide high-accuracy estimation.

According to **Fig. 4**, in GPR, the real function is the true, unknown function aimed to modelled, the estimated function is the GP's predicted mean representing the model's best guess of the real function, and the uncertainty is the predicted standard

deviation at each point, indicating how confident the model is in its estimate. For ripple loss estimation, uncertainty reflects how confident the model is about the predicted loss at a given operating condition, higher uncertainty indicates that the predicted ripple loss may deviate more from the real loss.

GPR can be fully defined by its mean $m(x) = Ex(l(x))$ and covariance functions $v(x, x') = Cov[l(x), l(x')]$, and performs well for small amounts of data [58], [32]. Here, $l(x)$ is a latent variable. Using these functions, GPR can be defined as shown in equations (8) to (10) [59]:

$$y = b(x)' + l(x) \quad (8)$$

$$l(x) \sim GP(0, v(x, x')) \quad (9)$$

$$b(x) \in \mathbb{R}^{p'} \quad (10)$$

where $b(x)'$ is the transpose of the deterministic basic function, p' is the dimension of the output and $v(x, x')$ is usually estimated as a hyperparameter that shows the level of linearity of the Gaussian process model. More information regarding GPR technique can be found in [57].

B. Decision Tree

DT is a non-parametric ML regressor using a hierarchical structure. It identifies patterns in inputs via leaf nodes and branches to produce outputs [60]. Common algorithms include Iterative Dichotomizer 3 (ID3) and C4.5 [61]. In ID3, entropy is minimized during the tree-building process, dividing data into regions based on optimal gains [60]. The entropy minimization and gain equations are shown in Equations (11) and (12) [61].

$$Entropy(S) = -P(yes) \times \log_2 P(yes) - P(no) \times \log_2 P(no) \quad (11)$$

$$Gain(S, A) = Et(s_i) - [W_{avg} * Et(s)] \quad (12)$$

where, $P(yes)$ and $P(no)$ denote the probabilities of 'yes' and 'no', s_i is the entropy of the i th sample, W_{avg} the weighted average, and $Et(s)$ the total entropy. Decision trees also use pruning to limit tree height, reducing overfitting from out-of-scope choices and invalid predictions [61].

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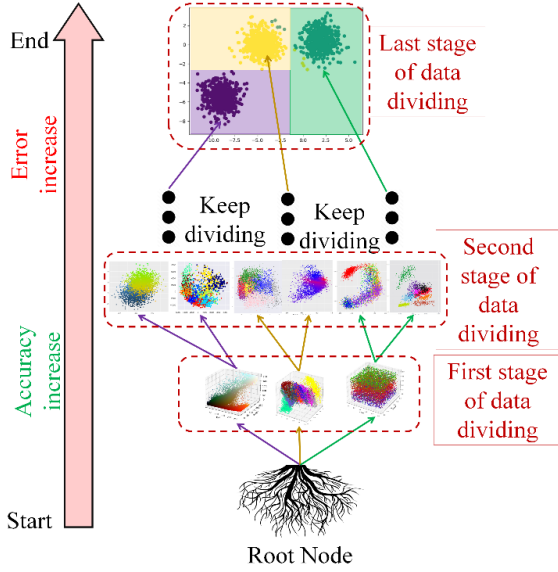


Fig. 5 An overview of DT structure for regression tasks

As depicted in **Fig. 5**, the decision tree algorithm stems from a root node, making a series of decisions as it moves downward through different layers in the tree structure, arriving, depending on the number of decision nodes, to a series of final ‘leaf’ nodes. More information about DT technique can be found in [60].

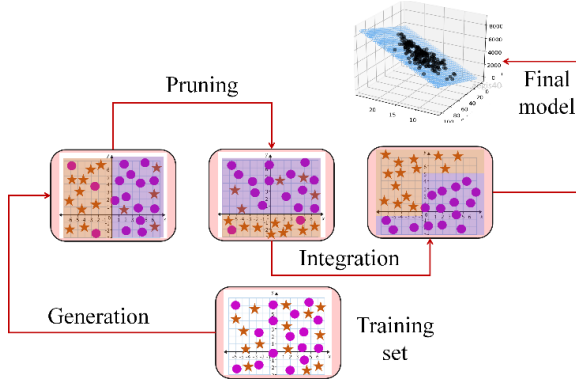


Fig. 6 An overview of ensemble learning process

C. Ensemble Learning

Ensemble learning is a supervised regression technique suitable for non-stationary scenarios [62]. It combines multiple models to improve numerical prediction accuracy [63], through three stages: generation, pruning, and integration [63]. In generation, models are created; pruning removes less-effective ones; integration combines the best models into a single ensemble. Skipping pruning (the “direct method”) can reduce accuracy, so it is included here, as shown in **Fig. 6** to improve ripple loss estimation [63].

Consider a dataset with n_{data} samples and m features, where their combination can be defined as shown in Equation (13) [64].

$$D = \{(x_i, y_i)\} \quad (|D| = n_{data}, x_i \in R^m, y_i \in R) \quad (13)$$

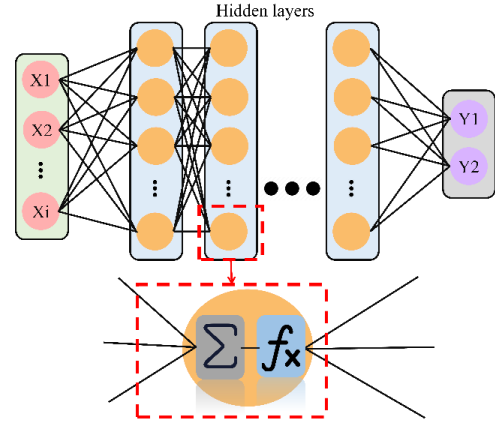


Fig. 7 Artificial neural network structure

The ensemble learning model φ is defined as shown in Equation (14) which is used to anticipate an output [64]:

$$\hat{y}_i = \varphi(x_i) = G(f_1, f_2, \dots, f_r) \quad (14)$$

where, G is aggregation function that aggregates r inducers for functions $\{f_1, f_2, \dots, f_r\}$.

The accuracy of the model is improved by a method known as weighting [63], which is based on adding the conditional probability vector obtained from each base model, as shown in Equation (15):

$$\hat{y} = \operatorname{argmax} (\varphi(x_i)), \text{ where } \varphi(x_i) = \sum_{c=1}^c f_r(x_i) \quad (15)$$

To define inducers there are different methods including Adaboost, bagging, least squared boosting, etc. [64].

D. Artificial Neural Network

Neural networks (NN), also called artificial neural networks (ANN), excel at learning complex behaviors. As shown in **Fig. 7**, inputs pass through hidden-layer nodes, each weighted and biased. Nodes compute a weighted sum and activate if it exceeds a threshold, passing outputs to the next layer [65]. The input–output dependency is shown in (16), and weights are updated to minimize the objective function in (17) [65].

$$y_{nn} = f_{act} \left(\sum_{i=1}^{n_{data}} \omega_i x_i + b_f \right) \quad (16)$$

$$F_{min} = Er_i = \sum_{i=1}^{n_{data}} y_{p_i} - y_{e_i} \quad (17)$$

where ω_i is the weight value, x_i is the input, b_f is the bias factor, f_{act} is the activation function, n_{data} is the number of data, Er_i is error value, y_{p_i} is the predicted output, and y_{e_i} is the expected output.

Various methods can update weights and biases. This study uses the backpropagation Levenberg–Marquardt (BPLM) method, proven effective in prior work. Equations (18)–(20) detail the BPLM update procedure [66]:

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$$\Delta w = (H_{es} + \mu_L I_{den})^{-1} g \quad (18)$$

$$g = J^T e \quad (19)$$

$$H_{es} = J_{bn}^T J_{bn} \quad (20)$$

where, Δw is the weight/bias change, H_{es} the Hessian matrix, μ_L the learning rate, I_{den} the identity matrix, J_{bn} the Jacobian, g the gradient, and e the error function. The Hessian, a square matrix of second-order partial derivatives, describes local function curvature and is widely used in calculus, optimization, and ML [66].

E. Performance Evaluation Metrics: Errors and Accuracy

To justify the accuracy and performance of the ML based intelligent regressors, Mean Squared Error (MSE), Relative Error (RE), Root Mean Squared Error (RMSE), fitting measure (R^2), Mean Relative Error (MRE), and Residual Value (RV) have been used in this paper as most common performance evaluation metrics, shown in Equations (21) to (26), [67]:

$$MSE = \frac{1}{n_{data}} \sum_{i=1}^{n_{data}} (y_{p_i} - y_{e_i})^2 \quad (21)$$

$$RE = \left| \frac{y_{p_i} - y_{e_i}}{y_{e_i}} \right| \times 100 \quad (22)$$

$$RMSE = \sqrt{MSE} \quad (23)$$

$$R^2 = \frac{\sum_{i=1}^{n_{data}} (y_{e_i} - \bar{y}_e)(y_{p_i} - \bar{y}_p)}{\sqrt{\sum_{i=1}^{n_{data}} (y_{e_i} - \bar{y}_e)^2 \sum_{i=1}^{n_{data}} (y_{p_i} - \bar{y}_p)^2}} \quad (24)$$

$$MRE = \text{mean}(RE) \quad (25)$$

$$RV = 1 - R^2 \quad (26)$$

where, $\bar{y}_e$ is the mean value of expected data and $\bar{y}_p$ is the mean value of estimated data.

Table 1 Parameters used in the study [46]

Parameter	Value
Tape width	4mm
ReBCO thickness	1 μ m
Copper stabilizer thickness	40 μ m
Substrate thickness	50 μ m
Silver thickness	2 μ m
Copper stabilizer conductivity at 77K	5.076x10 ⁸ Sm ⁻¹
Substrate conductivity at 77K	8x10 ⁵ Sm ⁻¹
Silver conductivity at 77K	3.704x10 ⁸ Sm ⁻¹
Permeability of free space	4 π x10 ⁻⁷ Hm ⁻¹
Power law exponent, n	38
Critical current, I_c	99.23A
Characteristic field ' E_c '	10 ⁻⁴ Vm ⁻¹

IV. RESULTS AND DISCUSSION

A. Model Validation

To validate the model, an HTS tape with 4 mm width was considered, based on [46]. The tape dimensions and parameters used in model are shown in **Table 1**. For loss validation, a purely sinusoidal current, without ripple, is injected into the HTS tape. Then, individual loss results in each sublayer were calculated, as shown in **Fig. 8**. The results show good agreement with the results of [46] and hence the model was considered validated.

Most of the loss is found within the ReBCO sublayers up to 1 kHz. In this region, the ReBCO sublayer has lower resistance, causing more current to flow into it. However, for frequencies more than 1 kHz and up to 85 kHz, skin effect impacts the current sharing.

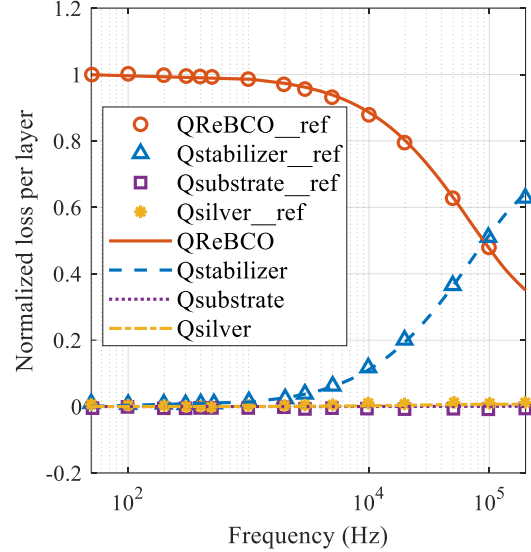


Fig. 8 Validation results for AC loss under a purely sinusoidal current, comparing to results from [46]

Table 2 Inputs and output range of the extracted data from the FEM model

Parameter	Min.	Max.	Mean	Standard Deviation	Median
Number of rectification poles	6	24	14	9.1652	12
$\frac{I_m}{I_c}$	0.1	0.8	0.45	0.2236	0.45
Frequency (kHz)	2	200	101	57591	101
Ripple loss (J/cycle/m)	5e-15	4e-5	3e-6	5e-6	9e-8

Thus, part of the current flows into metallic sublayers, especially the copper stabilizers. After 85 kHz, most of the loss is generated in copper stabilizers.

B. Data Preprocessing

shows the statistical distribution of the data extracted from the FEM model, where the number of poles, $\frac{I_m}{I_c}$ current amplitude, and frequency are the inputs for the ML based model while the loss value is the model output. Three different pole number for rectification are considered; 6, 12, and 24 pole. The ratio of operational current to critical current varies from 0.1 to 0.8 in steps of 0.05. The frequency varies from 2 kHz to 200 kHz in 1 kHz steps and the mean value shows the equal distribution of frequency for all data. The original dataset is divided into two subgroups, training data and test data. Training data is randomly selected from the original dataset to avoid any bias in ML model, comprising around 9000 data points based upon the output from the FEM simulation, where

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70% of the data is selected (training dataset). The remaining 30% of data is taken as test data. Test data are totally unseen by ML model during the training process to ensure robustness in the testing performance of the proposed ML based surrogate model. Therefore, during testing process these 30% of data will be treated as totally new data into the model.

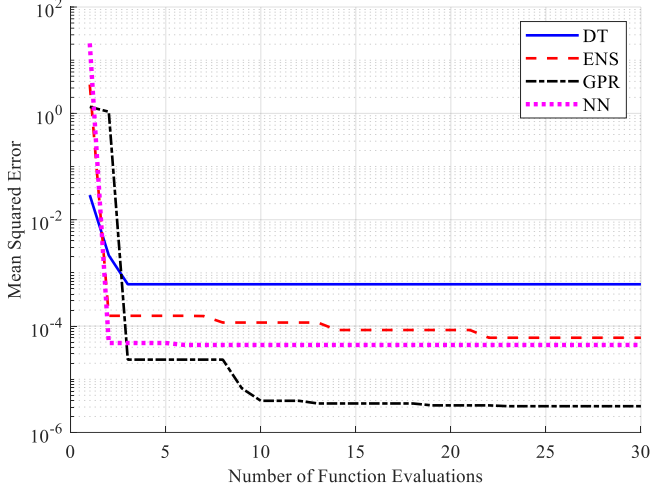


Fig. 9 Optimized cost function of each regressor based on number of evaluations

C. Hyperparameter optimization of ML Regressors

A global optimization has been conducted to find the best hyperparameters for each ML regressor. For the optimization purpose, a Bayesian optimizer has been used, based on the equations proposed in [68]. In literature, Bayesian optimizers have shown superiority and efficiency for hyperparameter tuning of ML methods[69, 70]. It has been shown that the Bayesian optimizer can achieve minimum error function in the least number of iterations. Unlike grid search, which exhaustively evaluates all combinations; or random search, which samples blindly; Bayesian optimization leverages probabilistic models and acquisition functions to intelligently explore the hyperparameter space. As a result, it achieves the minimum error in the few iterations by dynamically balancing exploration of uncertain regions and exploitation of known high-performance configurations. This makes Bayesian optimizer particularly advantageous for machine learning applications [71].

Fig. 9 shows the optimized cost function, which is a minimization of the MSE versus number of function evaluations. As shown in **Fig. 9**, GPR has the best performance of the regressors after optimization, with a minimum MSE value of $3e-6$, which is lower than the minimum MSE value of ANN, approximately 100 times lower than the minimum MSE of the ENS method, and more than 900 times lower than the minimum MSE of the DT method. The optimal hyperparameters as output of the performed optimizations are shown in **Table 3**. This will help the readers with further reproduction of our results in future.

D. Results of Optimized ML Regressors

After optimization of hyperparameters for the ML based intelligent estimators, they were tested to evaluate their accuracy. The results of the estimation by optimized regressors

are shown in **Fig. 10**, where the x-axis represents data extracted from the validated FEM based model and the y-axis shows the estimated data by ML based intelligent regressors.

The red trend line represents $y=x$, used as an index to show the fitness accuracy of the regressors; the closer the data points are

Table 3 Optimized hyperparameters of the regressors after going through the Bayesian optimization

Techniques	Optimized Structure
DT	Min parent: 10, Min leaf size: 2, Max split: 6267
ENS	Learn rate: 0.1402, learner type: Grown weak learners, boosting type: Least Squares Boosting (LSB)
GPR	Kernel function: Squared exponential, Basis function: constant, sigma: $1.4367e-04$
ANN	Layer size: [34 146 52], activation function: relu, lambda: $4.9702e-09$, layer weight initializing method: gloriot

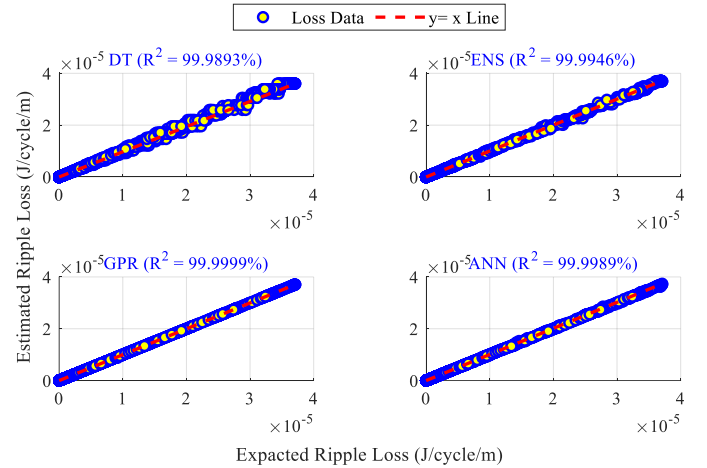


Fig. 10 Expected versus estimated ripple loss

to the $y=x$ line, the better the regressor performs and the higher the accuracy is. GPR has the best performance regarding the R^2 value, which is higher than all other techniques. The loss presented in **Fig. 10**, are related to loss in all sublayers.

Henceforth, the DT method, especially at higher loss values (see **Fig. 10**), performs poorly and with low accuracy. The ENS regressor has a better performance, compared to DT, however, for losses higher than $2e-5$ (J/cycle/m), there are still some fluctuations in accuracy. GPR and ANN regressors perform very well, showing a close agreement with real (expected) values.

To understand the accuracy of the ML based regressors in a quantitative manner, **Fig. 11** is presented showing the distribution of the estimated test data versus changes of frequency and current, where these loss values are related to total loss of HTS tape.

The maximum relative error of the DT method is about 2.5% which results in minimum accuracy of 97.5%. The minimum accuracy of the ENS regressor is about 98.6%, which is around 1.1% higher than the DT method. The ANN regressor suffers a

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maximum relative error of 4.5%, resulting in a minimum accuracy of 95.5%. Finally, the GPR regressor shows a minimum accuracy of 98%, which is close to DT, and less than the ENS regressor. Most data points in the GPR results are dark blue in color, which reflects that their relative error (of ML estimation) is less than 0.2%, and the maximum relative error

Table 4 Performance indices of optimized ML regressors, for test dataset

Techniques	RV (%)	RMSE (J/cycle/m)	MRE (%)	Train time (s)	Test time (s)
DT	0.011	0.018	0.228	2.571	0.051
ENS	0.003	0.009	0.107	2.982	0.059
GPR	0.003	0.003	0.012	10.943	0.139
NN	0.004	0.011	0.057	19.122	0.088

Table 5 Results of K-folding on different ML techniques, reported as mean MSE

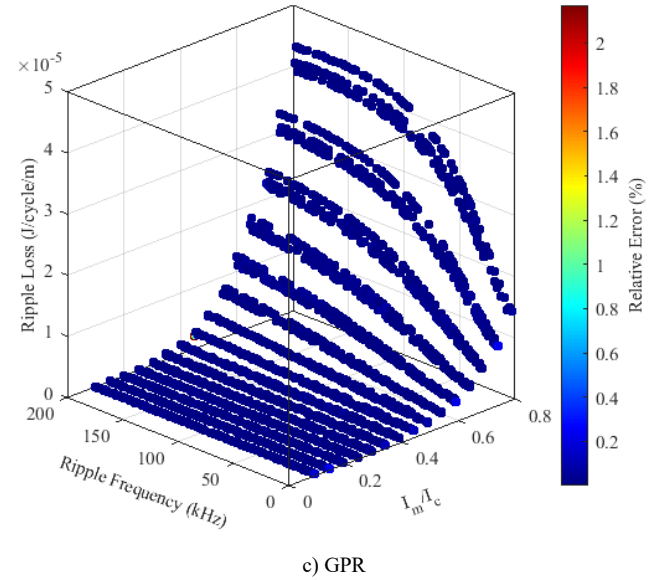
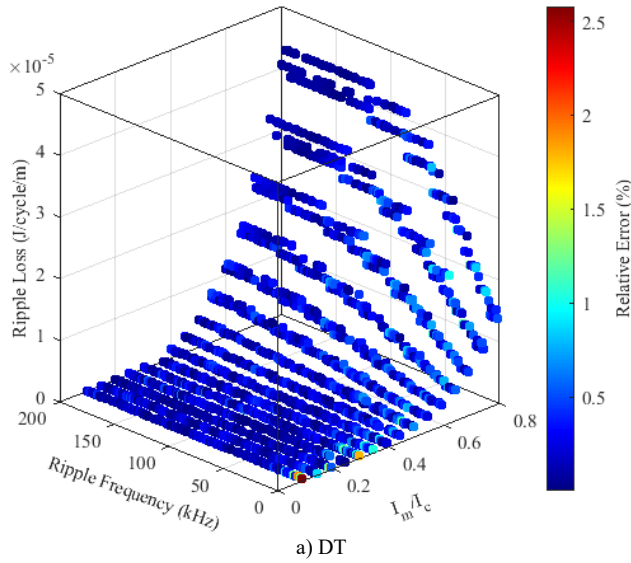
Loss Components	DT	ENS	GPR	ANN
ReBCO	5.7e-4	4.3e-4	5.8e-5	5.7e-5
Stabilizer	29e-4	7.9e-4	3.3e-4	3.8e-4
Substrate	30e-4	8.5e-4	6.4e-5	3.4e-4
Silver	53e-4	6.9e-4	4.5e-5	1.6e-4
Total	4.0e-4	1.6e-4	9.1e-5	4.6e-4

happens in only one point. Conversely, the ENS regressor has a lower maximum relative error, with the distribution showing a

lighter blue color for most points. This means that most of data points have a relative error higher than 0.5%. Hence, it could be concluded that GPR shows the best performance among other ML based regressors when considering overall accuracy as a selection index.

Table 4 shows the residual values, RMSE, MRE, training time, and test time of the optimized regressors. From **Table 4**, it is observed that, the RV value of GPR is lower than the ANN regressor, very close to ENS regressor, and much higher than DT method, hence showing GPR with the best performance. Considering RMSE, GPR shows the lowest value, i.e., the lowest error of ripple loss prediction is for the GPR regressor. For training time, GPR is higher than DT and ENS regressors. Additionally, test time of GPR is about 50% higher than ENS testing time and DT. However, considering training time as selection criteria, GPR is performing acceptably. MRE of GPR is 10 times smaller than ENS method, 5 times smaller than ANN, and 20 times smaller than DT. This shows that GPR is performing better than all other regressors, in terms of overall accuracy and error, as stated previously. Thus, it could be concluded from the results that GPR is the best ML based regressor studied for ripple loss estimation, using the available dataset.

Finally, to avoid overfitting, trained models are cross-validated, where cross-validation has been conducted based on the K-folding method, described in [72]. Generally, the idea behind K-folding cross-validation is dividing the original samples into “K” folds and the model is trained on “K-1” of the folds and validated on the remaining fold. This process is repeated “K” times, with each fold used exactly once as the validation data, hence guaranteeing the proposed model is not overfitted. In literature, it is noted that 5 folds is a good measure for ML algorithms with low numbers of data in the dataset [73, 74].



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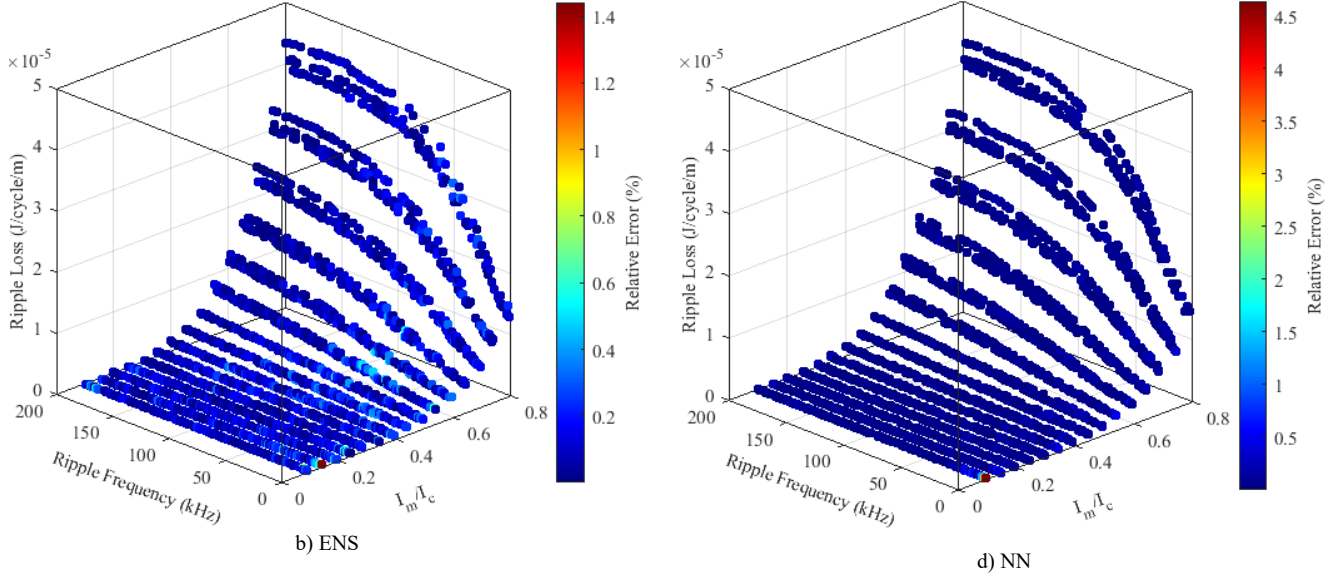


Fig. 11 Relative error distribution and values for each optimized regressor, test data results

Table 5 illustrates results of K-folding with $K = 5$ for different components of ripple loss. The table shows MSE for all folds. From **Table 5**, the best performance of cross-validation relates to GPR with the mean squared loss value less than $4e-4$ for all components. Among different loss components of folds, the ReBCO sublayer-related loss shows the lower MSE. The main takeaway of this table is that the proposed ML based regressors are not overfitted. Upscaling AI-based models may be conducted by transfer learning, [75, 76]; hybrid modelling [77, 78], or surrogate-based system modelling [79].

E. Generalization of Best Performing Regressor

To evaluate the generalizability of the GPR model, additional studies were conducted by altering both the geometry and the size of the training dataset, and the operating temperature range of the ReBCO tape. Firstly, the geometry of the model was modified to include HTS tapes with widths of 10 mm and 12 mm, which are commonly used in practice, to examine whether the model can reliably predict losses for tapes of different dimensions. Furthermore, to test the model's dependence on data quantity, the training dataset was reduced to approximately 10% of its original size (about 900 data points) by limiting the number of frequency points and transporting current levels considered in the parameter sweep.

Table 6: Ripple estimation error for extrapolation performance of GPR

Type of Extrapolation	Unseen 12 mm	Re-trained 12 mm	Unseen 10 mm	Re-trained 10 mm
R^2	0.964	0.975	0.967	0.976
RV	0.036	0.025	0.033	0.024
MRE (%)	5.125	4.127	4.520	3.579
Silver	53e-4	6.9e-4	4.5e-5	1.6e-4

Total	4.0e-4	1.6e-4	9.1e-5	4.6e-4
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Two approaches were adopted for this evaluation. In the first approach, referred to as the “Unseen” case, data from the 12 mm and 10 mm HTS tapes were used directly as test data for the trained GPR model to assess its predictive capability on entirely unseen geometries. In the second approach, referred to as the Retrained case, the model was retrained using only the reduced dataset of 900 data points. The performance of both approaches was compared against FEM simulation results. As summarized in **Table 6**, the GPR model maintained a high level of accuracy for both unseen and retrained scenarios, with small relative errors and strong goodness-of-fit values. **Fig. 11** further illustrates the close agreement between the expected FEM results and the GPR-predicted losses for the 12 mm and 10 mm tapes under various operating conditions.

These findings confirm that the proposed model effectively captures the underlying physical relationships governing ripple losses rather than relying on memorized patterns from training data. Even when applied to tape geometries above the original training set, the model demonstrated consistent predictive performance. Retraining the model on 10% of the original dataset further improved accuracy, highlighting its efficiency and adaptability.

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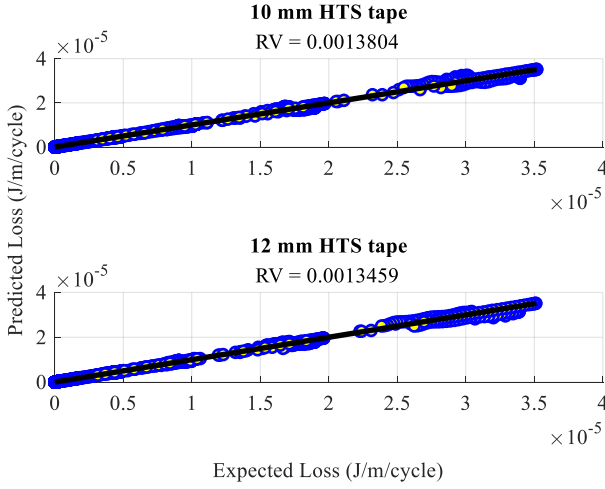


Fig. 12: Estimated loss by extrapolation performance of GPR

To test generalizability at different operating temperatures, the 4 mm tape FEM model was re-run at temperatures in the range 40 K to 60 K, with the data then fed into the GPR model and re-trained on the new dataset. The output of the GPR model is shown in **Fig 12** for selected temperatures, with the GPR regressor showing good agreement in the majority of cases, demonstrating the adaptability of the proposed ML model to different temperatures, and showing the generalizability of the proposed regression model.

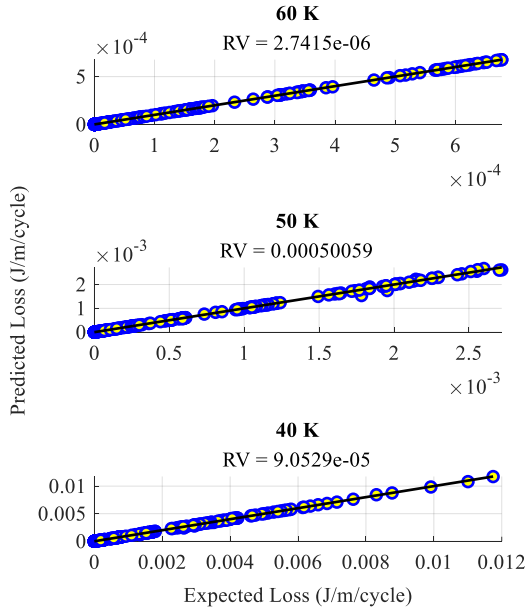
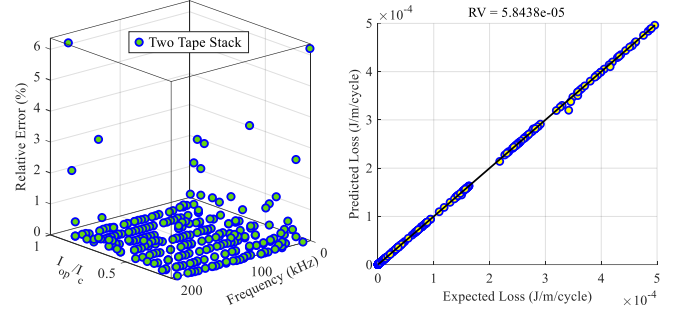


Fig. 13: Performance of GPR model for estimating loss at different temperatures

F. Upscaling to a two-tape stack model

To further clarify the generalizability of the model to other geometries, a two-tape ReBCO stack was modelled in COMSOL using the same geometric parameters as described previously for the 4 mm width. Stacks are typically used as the next step up from single tape level geometry and have been considered for use in electric propulsion previously in [80].

The FEM model was run for 900 cases, 10% of initial training data, and the new data was used to retrain the GPR model, where the results can be observed in **Fig. 14**, where the predicted loss from the ML model shows good correlation with the expected loss obtained from the FEM simulations. The relative error is also demonstrably low for the vast majority of data points, at below 1%, with only a couple of outliers at around 6%.



a) Relative error for two-tape stack

b) Expected versus estimated ripple loss

Fig. 14 Performance of the GPR model for estimating stack level results

In this study, the ML models used fall at tape and small scale stack level. It would be reasonable to upscale any of these models to consider further geometries, such as a superconducting coil or cable, however, both the FEM and AI models would need alteration to account for the different loss mechanisms at more complex geometries, such as the rotating field of a superconducting machine.

V. CONCLUSION

Ripple currents, arising from AC/DC conversion introduced additional AC-like losses in HTS tapes are a notable issue for the drivetrains or cryo-electric aircraft. This paper proposes data-driven machine learning (ML) approaches for the rapid and accurate prediction of such ripple losses. Loss data extracted from a validated FEM model was used to train four different ML regressor algorithms, GPR, DT, ENS, and ANN. The input features used for these models were the number of rectification poles (pulses), ripple frequency, and current ratio. The main findings of this study are as follows:

- The best performing ML-technique was GPR which showed the highest mean accuracy (99.98%) and lowest relative squared errors in prediction, about 0.003 (J/cycle/m)
- The second-best ML-technique was ANN, then ENS, and then DT
- The fitness accuracy for all 4 ML based regressors showed a very promising result; above 98%
- Train time of ML-techniques ranged between 2.5 s to 20 s, compared to FEM simulations taking 10–30 minutes, per simulation
- Test time of ML-techniques ranged between 51 ms to 139 ms, compared to FEM simulations taking 10–30 minutes, per simulation
- The proposed GPR regressor showed good generalizability, performing well for increased tape geometry, different operating temperatures, and for a two-tape stack model

The results have demonstrated the computational efficiency and accuracy of ML-based techniques for ripple loss estimation when compared to the far lengthier time required for FEM based models to be continually updated and re-evaluated. The developed tape-level models, both physics and AI-based, form a foundation for multi-scale studies from stacks and coils to entire drivetrains, capturing essential electromagnetic and loss characteristics of HTS tapes that allow upscaling of the model without retraining.

Thus, the proposed ML framework can facilitate rapid design optimization and real-time system analysis. This study can also be extended for device-level analysis of HTS machines, or even further, for powertrain-level for heat and loss estimation in cryo-electric aircraft, as these flying units enable energy decarbonization of the industry.

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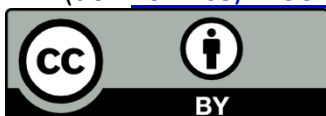
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